Adam: Adaptive Moment Estimation

Optimization problem

We want to minimize (or maximize) an objective

$$\min_{\theta \in \mathbb{R}^d} f(\theta),$$

typically of the form

$$f(\theta) = \mathbb{E}_\xi[\,\ell(\theta;\xi)\,] \quad \text{or} \quad \frac{1}{n}\sum_{i=1}^{n} \ell(\theta;i).$$

Common in:

- Machine learning and deep learning
- Large-scale stochastic optimization
- Nonconvex problems

From gradient descent to stochastic gradients

Deterministic gradient descent:

$$\theta_{t+1} = \theta_t - \eta_t \nabla f(\theta_t).$$

Stochastic gradient descent (SGD):

$$\theta_{t+1} = \theta_t - \eta_t g_t, \qquad \mathbb{E}[g_t \mid \theta_t] = \nabla f(\theta_t).$$

Issues with SGD:

- Noisy gradients
- Sensitive to learning-rate choice
- Same step size for all coordinates

Motivation for Adam

Adam addresses two key issues in SGD:

- **Noise**: smooth gradients using momentum
- **Scaling**: adapt step sizes per coordinate

Core ideas:

- Track first moment (mean) of gradients
- Track second moment (variance) of gradients
- Normalize updates using both

Adam as a statistical estimator

Reframe: Adam is a **method of moments estimator** applied online.

At each step t, Adam maintains running estimates of the gradient distribution:

$$m_t \approx \mathbb{E}[g_t], \qquad v_t \approx \mathbb{E}[g_t \odot g_t].$$

The update is a **normalized score direction**:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \approx \theta_t - \eta \cdot \frac{\mathbb{E}[g]}{\sqrt{\mathbb{E}[g^2]}}.$$

Key insight: The denominator $\sqrt{\hat{v}_t}$ approximates the *standard deviation* of past gradients per coordinate — a form of **online whitening**.

This is why Adam is robust: it standardizes the gradient signal coordinate-wise at every step.

First and second moments

Given stochastic gradient g_t:
First moment (momentum):

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

Second moment (RMS scaling):

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t \odot g_t$$

where:

- $\beta_1 \in [0, 1)$ controls momentum
- $\beta_2 \in [0, 1)$ controls variance smoothing
- $\odot$ denotes elementwise square

Bias correction

Since $m_0 = v_0 = 0$, early estimates are biased toward zero.
Bias-corrected moments:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \qquad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}.$$

This correction is important in early iterations when t is small.

Adam update rule

Adam parameter update:

$$\boxed{\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}}$$

Components:

- numerator: smoothed gradient direction
- denominator: coordinate-wise scaling
- $\epsilon > 0$: numerical stability

Adam algorithm (plain pseudocode)

Input: step size η, parameters $\beta_1, \beta_2, \epsilon$

Initialize: θ_1, $m_0 = 0$, $v_0 = 0$, $t = 0$

Repeat:

- $t \leftarrow t + 1$
- Sample mini-batch and compute stochastic gradient g_t
- Update moments:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t \odot g_t$$

- Bias correction:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

- Update parameters:

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

Output: final or averaged iterate

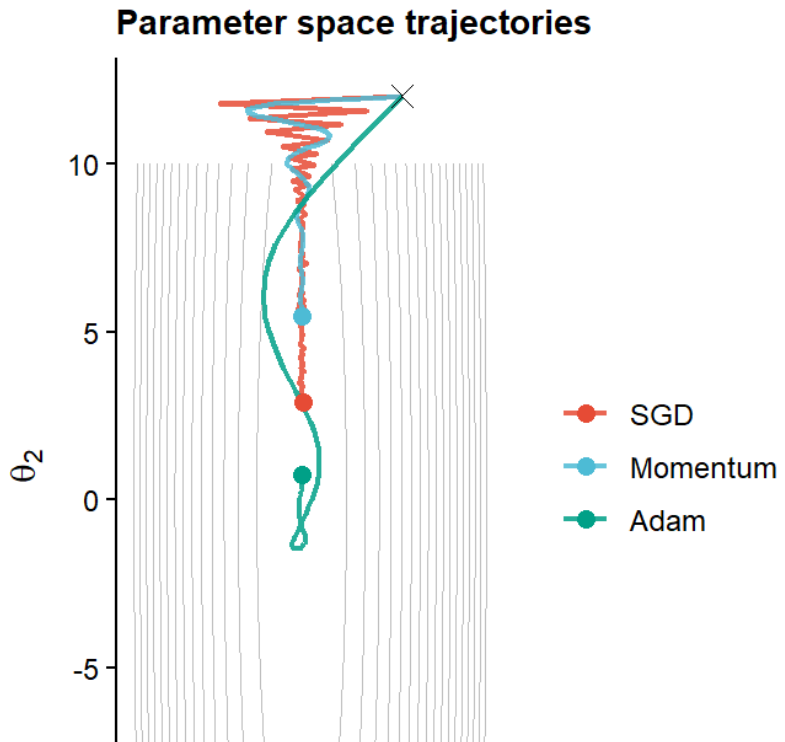
Interpretation of Adam

Adam combines:

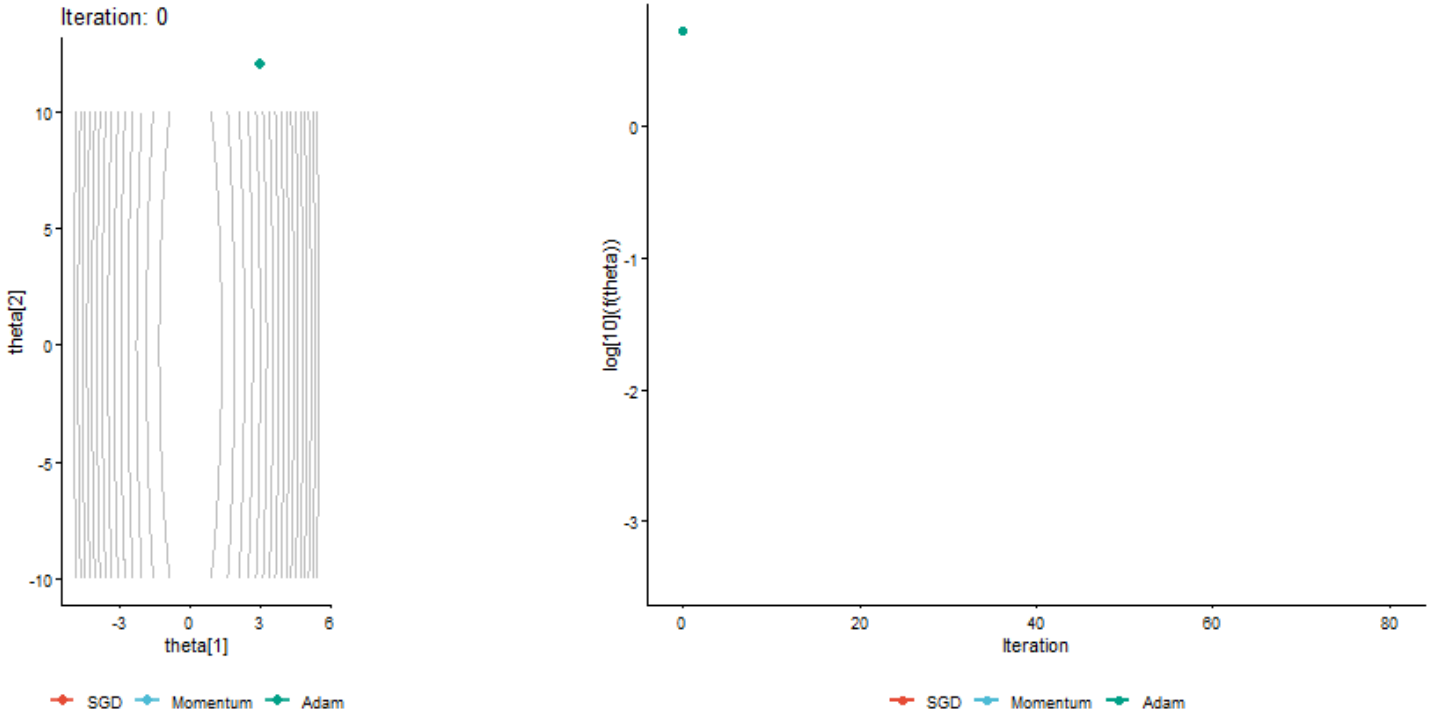
- **Momentum** (like heavy-ball / Polyak)
- **Adaptive scaling** (like AdaGrad / RMSProp)

Effects:

- Large gradients $\Rightarrow$ smaller effective steps
- Small gradients $\Rightarrow$ larger effective steps
- Reduced sensitivity to learning-rate tuning



Comparison: SGD, Momentum, Adam



Left: parameter-space trajectories. Right: loss vs. iteration.

Typical hyperparameters

Common default values:

$$\beta_1 = 0.9, \qquad \beta_2 = 0.999, \qquad \epsilon = 10^{-8}.$$

Learning rate:

$$\eta \in [10^{-4}, 10^{-2}] \quad \text{(problem dependent)}.$$

Defaults work surprisingly well, but tuning can still matter.

Convergence considerations

Adam performs well empirically, but theory is subtle.
Key points:

- Original Adam may fail to converge in some convex settings
- Modified variants (e.g. AMSGrad) restore convergence guarantees
- In practice, Adam converges reliably for many nonconvex problems

Theory typically shows convergence to stationary points under assumptions.

Practical tips

- Combine Adam with learning-rate decay or warmup
- Use gradient clipping for stability
- Monitor effective step size $\|\eta \hat{m}_t / \sqrt{\hat{v}_t}\|$
- For final convergence, sometimes switch to SGD

Adam is excellent for fast progress, not always for final precision.

Common pitfalls

- Too large $\eta \Rightarrow$ instability despite adaptivity
- Poor generalization in some settings compared to SGD
- Over-reliance on defaults without diagnostics
- Ignoring convergence criteria (gradient norms)

Adam is powerful, but not magic.

Connection to weak lensing

Adam is the optimizer in **ShearNet**, a neural network for weak gravitational lensing shear estimation.

Why Adam fits naturally:

- Noisy gradients: galaxy images are shot-noise limited
- Anisotropic loss landscape: shear components (γ_1, γ_2) and PSF parameters operate on different scales
- Coordinate-wise adaptivity handles the heterogeneous parameter space automatically

The statistical framing is literal here: m_t and v_t track the empirical gradient distribution over a stochastic mini-batch of simulated galaxy observations.

This is a direct application of online moment estimation to an astronomical inverse problem.

Takeaways

- Adam adapts step sizes using first and second moments of gradients
- Combines momentum and RMS normalization
- Highly effective for large-scale, noisy, nonconvex optimization
- Default optimizer in many modern ML systems

Adam: adaptive, robust, scalable stochastic optimization